

Customization Quick Reference — Precision Flow Conditioner

Expected Effects of Parameter Changes (Qualitative Guidance Only)

Parameter	Direction	Main Effect	Secondary Effects	Practical Notes
Gap size	↓ 0.6–0.8 mm	Higher velocity & shear	Higher Re, smaller bubbles	Harder to print & assemble consistently
Gap size	↑ 1.4–1.8 mm	Gentler flow	Lower Re, larger bubbles	Easiest to manufacture
Nozzle angle	↓ 6–8°	Stronger/longer spiral	Better conditioning, ↑ΔP	Best uniformity
Nozzle angle	↑ 12–15°	Shorter path	Easier high flow, less conditioning	Good for throughput
HVAT brake	30–50% locked	Major swirl reduction	Clean axial jet	Most powerful tuning knob
# of discs	↑ 30–35+	More parallel channels	Smoother output, taller device	More print time & material
Orifice dia	↓ 6 mm	Highest jet velocity	Most violent collapse (low σ)	Risk of excessive erosion

DISCLAIMER: These are qualitative expectations derived from simplified boundary-layer and cavitation models. They are for hypothesis generation and educational purposes only. Real-world results depend on surface finish, alignment precision, inlet conditions, fluid properties, and many other variables. Always validate experimentally.